LLM-Driven Personal Trainer Matching Based on Client Goals, Body Metrics, and Availability

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***Abstract*—Matching clients with personal trainers requires satisfying hard requirements such as training goal, available hours, budget, training mode and district, and ordering acceptable trainers by fit. We study how a large language model can be combined with deterministic constraint filtering and multi-criteria compatibility scoring in the Gymini fitness platform. Validity, ranking quality and explanation accuracy are evaluated separately in three experiments: an offline simulation with 80 or 250 trainers and 300 clients over five seeds; a benchmark of 20 publicly referenced trainer profiles with synthetic operational attributes and 30 synthetic clients; and 138 real calls to Claude Opus 5.5 under three configurations that differ only in the candidate list. In simulation, hard filtering reduced constraint violations among the top five from about 95% to 0%, whereas the production compatibility score ranked below a filter-then-rating baseline in all four scenarios. On the benchmark, 26.9% of the trainers returned by the current pipeline were valid, mainly because availability is stored per day and location per province. The language model alone produced no constraint-violating recommendation in 30 test cases (95% confidence interval 88.6% to 100%), and, under test conditions, the filtered configuration reached the same validity at about 57% lower cost. Ranking quality could not be distinguished from random ordering of valid trainers because of a ceiling effect. Free-text explanations copied every numeric fact correctly but overstated simulated trainer traits in 23.9% of cases; a claim catalog removed model-added content, yet its fixed templates asserted unverified experience in 23 of 27 explanations.**

***Index Terms*—Constraint-based recommendation, explainable recommendation, large language models, multi-criteria recommender systems, personal trainer matching, synthetic data.**

I. Introduction

PERSONAL trainer platforms must match each client with trainers who satisfy hard requirements, such as training goal, available hours, budget, training mode and location, before ordering the acceptable trainers by fit. The task resembles people-to-people recommendation, in which a suggestion is useful only when it is acceptable to both sides [1], and it involves several criteria that a single overall rating does not capture [2]. Constraint-based recommenders handle explicit requirements by checking items against rules [3], while large language models (LLMs) are increasingly used to rank items and to explain recommendations in natural language [4]–[6].

Using an LLM in this role raises three separate questions. First, the model may recommend trainers that violate stated requirements or do not exist in the catalog, a form of hallucination [7], and LLM rankers are sensitive to the order in which candidates appear in the prompt [6]. Second, a high ranking metric does not by itself show that the order among acceptable trainers is good. Third, a fluent explanation can assert facts that the data do not support [7], [8]. Grounding the model in retrieved or structured data is a common mitigation [9], but its effect on each question has to be measured separately.

We study these questions in Gymini, a fitness platform in which a conversational LLM collects the client's criteria, a hard-constraint filter and a multi-criteria compatibility score (scorePT) select and order trainers, and a claim catalog constrains the explanations. The contributions are: (i) an evaluation design that keeps validity, ranking quality and explanation accuracy apart across three experiments, namely an offline simulation, a benchmark of 20 trainers and 30 clients, and 138 real calls to Claude Opus 5.5; (ii) an explicit data design that separates real public trainer information, AI-generated synthetic attributes and code-generated simulation data field by field, with the real part pseudonymized; and (iii) measured findings, including negative ones, on where filtering, scoring and the LLM help and where they do not. Section II reviews related work, Section III describes the methodology and data, Section IV reports the results, Section V discusses them and Section VI concludes.

II. Related Work

Multi-criteria and constraint-based recommendation. Adomavicius and Kwon proposed recommendation techniques that exploit ratings on several criteria instead of a single overall rating [2]. Constraint-based recommenders encode user requirements and item properties as constraints and return only items that satisfy them, which also raises the question of what to do when no item does [3]. Matching clients with trainers is in addition a people-to-people problem: reciprocal recommenders such as RECON take the preferences of both sides into account [1].

LLMs as recommenders. Wu *et al.* survey the use of LLMs in recommendation, either as components of conventional pipelines or as recommenders themselves [4]. Liu *et al.* evaluated ChatGPT on several recommendation tasks, including the generation of explanations [5], and Hou *et al.* showed that LLMs can rank candidates zero-shot but are sensitive to candidate position and favor popular items [6].

Hallucination, grounding and explanation. Ji *et al.* survey

hallucination, i.e., generated content that is nonsensical or unfaithful to the provided source [7]. Retrieval-augmented generation conditions the model on retrieved documents for knowledge-intensive tasks [9], and explainable recommendation studies how to justify recommendations to users [8]. These works mostly evaluate ranking accuracy or explanation quality on public data. We are not aware of a public benchmark for personal trainer matching, and the present work therefore reports validity, ranking quality and the provenance of explanations separately, together with a declaration of how its evaluation data were produced.

III. Research Methodology

A. Research Questions

We address four research questions. RQ1: how do hard constraints on goal, availability, budget, training mode and district affect the validity of recommended trainers? RQ2: after filtering, does multi-criteria compatibility scoring, which includes body metrics through the history of similar clients, improve ranking quality over simple baselines, and which criteria contribute? RQ3: with a real LLM, how does letting the model select trainers on its own compare with combining it with filtering and scoring in validity, ranking quality, fabricated trainers, stability and cost? RQ4: are explanations written freely by the LLM, and explanations assembled by a claim catalog, faithful to the data and to the verification level of their sources?

B. System Under Study

Gymini organizes trainer matching into four tiers (Fig. 1). T1 is a conversational LLM that collects the client's criteria over several turns; it is described but not evaluated. T2 is a hard-constraint filter that keeps trainers who are approved, still accepting clients and hold the required specialty, are available on every day the client selected, and offer a package within budget, a suitable session length and enough free slots in the next 28 days. T3 is the compatibility score scorePT (version compatibility-v2):

$$s = \mathrm{round}\left(100 \cdot \frac{\sum_{k \in K} w_k c_k}{\sum_{k \in K} w_k}\right) \quad (1)$$

where $c_k \in [0, 1]$ is the value of criterion k, w_k its weight and K the set of criteria in use. The weights are goal 30, schedule 25, budget 20, reputation 10 and evidence 15. Goal checks whether the trainer's specialty matches the client goal, schedule is the share of the client's days on which the trainer is available, budget checks whether a package fits the budget, and reputation is the normalized rating. Body metrics enter through evidence, which counts past clients of the trainer whose body-metric profile is similar to the current client (similarity ≥ 0.65); it requires at least five such clients and saturates at 20. With fewer than five, evidence is removed from both sums in (1), so a new trainer is not penalized for missing history. The score is rounded to an integer and ties are broken by trainer identifier. T4 is a claim catalog: the system builds candidate claims from data fields, the LLM selects only claim identifiers, the sentences come from fixed templates, and unknown identifiers are discarded. The claim catalog is existing Gymini code and is not built from external sources.

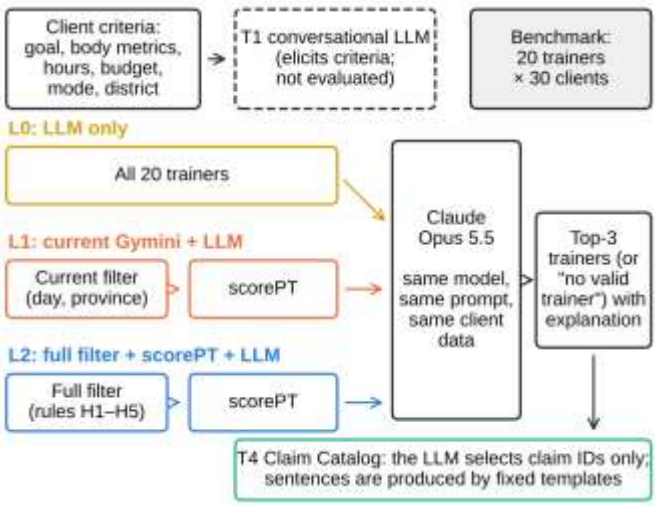


Fig. 1. Matching pipeline and the three LLM configurations of Experiment B, which share the model, system prompt, parameters and client data and differ only in the candidate list.

C. Experimental Data Design

Because the evaluation mixes real and synthetic data, every data element is labeled with its origin (Table I). The label is also stored in the data files: each trainer record has a profile_data_kind flag marking its attributes as synthetic with a public profile reference, and each client record has is_synthetic = true and a body-fat value marked as simulated, not as a clinical measurement.

TABLE I
ORIGIN OF THE EXPERIMENTAL DATA

Data element	Origin	Generation and verification
20 trainer profiles and their gym websites	Real, public	Five public gym websites in Ho Chi Minh City; names and URLs withheld
Specialty labels	Real, normalized	11/20 match public statements exactly; 4 set to GENERAL_FITNESS (none stated), 4 inferred from descriptions, 1 with an omission
Experience, price per session, mode, districts, weekly slots, coaching traits	Synthetic (AI-generated)	Generated with Claude and reviewed by the authors; only 2/20 experience values match public claims
30 client scenarios	Synthetic (AI-generated)	Generated with Claude, reviewed and locked; precheck 30/30
Simulation of Experiment A	Synthetic (code)	Generated deterministically from code with seeds 11, 23, 37, 41, 53
Validity and soft labels	Rule-based	Computed by the authors' rules; soft labels frozen before the first LLM call

Real public data. The benchmark (snapshot GYMINI_SIM_COMPLETE_V2_2026-11-03) contains 20 trainer profiles referenced from five public gym websites in Ho Chi Minh City. Specialty labels were normalized from public statements when available; the remaining labels were assigned or inferred. Public statements were not independently verified, and the source ledger records every claim as not independently verified.

AI-generated synthetic data. Operational attributes that the websites do not publish (price per session, availability, district, training mode, experience and coaching traits) and all 30 client scenarios were generated with Claude (Anthropic) [10], reviewed by the authors and locked before the experiments (status LOCKED_FOR_BENCHMARK_SIM_V2). The 20 trainers offer 26 packages (20 offline, 6 online) priced from 240,000 to 520,000 VND per 60-minute session, with five or six weekly time slots per trainer, in nine districts or online. The

clients are aged 21 to 43 (15 female, 15 male; 12 beginner, 10 intermediate, 8 returning), have a body-fat percentage of 13.5% to 30.8%, one of seven goal codes, available hours per day, a budget of 250,000 to 600,000 VND per session or no limit (3 clients), a training mode (20 offline, 6 online, 4 hybrid), a district and one soft preference; the specialty, budget and location requirements are each flagged as hard or soft. The scenarios follow a coverage design: each client has a designated coverage trainer that satisfies all of its hard constraints, as confirmed by a precheck, so that every client has at least one valid trainer. As a result 18 clients have exactly one valid trainer, 12 have two or more, and none has zero.

Labels. A recommendation is valid when it satisfies six rules: H0, the trainer code is one of the 20 trainers; H1, the trainer has every required specialty; H2, there is at least one overlapping slot of at least 60 min on the same day; H3, the price per session is within budget; H4, the training mode matches; and H5, the district matches for offline training with a required location. Price, mode and district must be satisfied by the same package. The labels are computed by these rules, not by the generator, and agree with the independent precheck for 30 of 30 clients. For ranking within valid trainers we use soft labels frozen at 09:56:11Z on 7 November 2026, before the first LLM call at 09:57:23Z: a valid trainer has gain = 1 + 0.5·softPref + 0.5·goalPrimary, where softPref = 1 if the client's soft preference matches the trainer's coaching traits by keyword and goalPrimary = 1 if the client's primary goal is among the trainer's specialties, so the gain takes only the values 1, 1.5 and 2; an invalid trainer has gain 0.

Simulation (Experiment A). The simulation generates 80 or 250 trainers and 300 clients per seed over five seeds. Each trainer has a latent quality q per goal and body-type group (lean, average, high body fat); client histories and star ratings are generated from q with noise. The gain equals q for a feasible trainer and 0 otherwise, and a trainer is relevant when its gain is at least 0.6. Scenarios S1 and S2 use 80 trainers and S3 and S4 use 250; ratings reflect quality in S1 and S3 and are inflated in S2 and S4.

Protection of real data. No data about real clients were collected. Because names are personal data under Vietnamese law [11], trainers appear only as PT001–PT020 and websites as W1–W5. In the artifact package shared for review, the free-text public statements are replaced by keyed HMAC-SHA256 digests, so the authors can check that a statement is unchanged without disclosing it; the key, the code mapping and the original files are kept in a separate private folder that is not shared. A scan of the shared package found no website name. The benchmark files contain no trainer names or profile URLs.

D. Evaluation Design

The evaluation is organized by aspect rather than by experiment (Table II): each aspect has its own metrics and its own reference point, and a result on one aspect is never used as evidence for another.

TABLE II
EVALUATION ASPECTS, DATA, METRICS AND REFERENCES

Aspect	Exp.	Metrics	Reference points
Validity	A, B, C	Violation@5; Valid@3, Top-1 validity, case success, recall, empty answers, fabricated codes	Random; rating only; oracle floor
Ranking quality	A, B	NDCG@5, HitRate@1 (A); NDCG@3 on 12 clients (B)	Oracle; ablations; random valid ranking; worst valid order
Explanation accuracy	B	Numeric facts; (a) overclaims; (b) unsupported verification claims; claim-ID validity	Free text vs. claim catalog
Stability	B	Identical top three; Jaccard index	Second run of 10 clients
Cost and latency	B	Tokens, latency, USD per 1000 calls	L0 vs. L1 vs. L2

E. Compared Methods

Experiment A compares M0 random ordering, M1 rating only, M2 scoring without filtering, M3 filtering then cheapest, M4 filtering then rating, M5 filtering then scorePT v1 and M6 filtering then scorePT v2, the production method. Ablations A1–A3 remove goal, schedule and budget together, reputation, and body-metric evidence; variant X2 (no rounding, ties broken by rating) is an analysis only. An oracle that ranks by true gain reaches NDCG@5 = 1.000 and HitRate@1 = 1.000, which checks the metric implementation. *Experiment B* runs three configurations that share the system prompt, model, parameters and client data and differ only in the candidate list: L0 (LLM only) receives all 20 trainers; L1 (current Gymini + LLM) receives the trainers returned by G0, the current filter followed by scorePT, with their scores; and L2 (full constraints + scorePT + LLM) receives the trainers valid under H1–H5 ranked by scorePT (G1), with their scores. An audit confirmed that the L2 list equals the valid set for all 30 clients (1.73 trainers on average). The LLM must check the constraints, return at most three trainers or NO_VALID_PT, and answer in JSON with the chosen package, the overlapping slot and an explanation. Non-LLM references are G0, G1 (top three) and a random valid ranking that shuffles each client's valid trainers over 1000 seeds. *Experiment C* evaluates the Gymini pipeline itself: R, three of the 20 trainers drawn at random (mean of 200 draws); G0, the current pipeline; and G1, G0 extended with hour, district and per-session price constraints.

F. Metrics and Statistical Analysis

Valid@3 is the number of valid positions divided by the number of positions returned; Coverage@3 is the number of positions returned divided by three times the number of clients; Top-1 validity is the share of clients whose first trainer is valid (an empty answer counts as a failure); case success is the share of clients with at least one valid trainer in the top three; and recall is $|\text{top-3} \cap V| \,/\, \min(3, |V|)$, where V is the valid set. Ranking quality is measured by

$$\text{NDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k}, \quad \text{DCG@}k = \sum_{i=1}^{k} \frac{2^{g_i}-1}{\log_2(i+1)} \tag{2}$$

where g_i is the gain at rank i and IDCG@k uses the gains sorted in decreasing order [12]; ties need no tie-breaking because IDCG depends only on gain values. Experiment A uses $k = 5$ with gain q, and Experiment B uses $k = 3$ with the soft labels on the 12 clients that have at least two valid trainers. In Experiment A, HitRate@1 is the share of queries whose first trainer has the

highest gain among feasible trainers, counting ties, and Violation@5 is the share of returned trainers that are infeasible, with a floor equal to the violation rate of the oracle when it must always return five trainers. For explanations we check four numeric facts per recommendation (mode, price, district and a shared slot of at least 60 min) and count two error types separately: (a) *overclaim*, detected by regular expressions (e.g., rated, reviewed, certificate, guarantee) followed by manual reading of every flagged sentence; and (b) *unsupported verification claim*, stating that an attribute is verified or approved when the benchmark marks it as simulated or unverified. Proportions use 95% Wilson intervals [13]; NDCG and paired differences use a bootstrap over clients with 10,000 resamples and seed 2026 [14]; Experiment A uses paired t intervals over the five seeds (df = 4). Cost is input tokens × 4 USD per million plus output tokens, including reasoning tokens, × 20 USD per million.

G. Experimental Protocol

All LLM calls used Claude Opus 5.5 (claude-opus-5-5) [10] through the Anthropic Messages API with max_tokens = 16000, the default reasoning level, no sampling parameters (not supported by the model) and no fallback model. We ran 90 main calls (30 clients × 3 configurations), 30 repeated calls (the first 10 clients with at least two valid trainers × 3 configurations, chosen before any result was inspected) and 18 claim-catalog calls (clients C01–C18): 138 calls in total, with 0 errors, every stop reason end_turn, and a total cost of 4.77 USD. Each call stores the request, raw response, token usage, latency and hashes of the prompt and data. An independent audit recomputed every metric of Experiment B in Python from the raw responses, and re-running Experiments A and C, the scoring of B and the confidence intervals reproduced the reported numbers exactly, apart from run-time measurements.

IV. Experiments and Results

A. Experiment A: Filtering and Ranking at Scale

TABLE III
FILTERING AND RANKING IN SIMULATION (MEAN ± SD OVER 5 SEEDS)

Method	NDCG@5 S1	NDCG@5 S4	Hit@1 S1	Hit@1 S4	Viol@5 S1	Viol@5 S3
M0 Random	0.054±0.006	0.038±0.007	0.011	0.001	94.9%	94.7%
M1 Rating only	0.049±0.027	0.038±0.017	0.025	0.000	95.9%	94.9%
M2 Score, no filter	0.598±0.097	0.538±0.029	0.211	0.147	49.6%	37.8%
M3 Filter + cheapest	0.887±0.022	0.776±0.022	0.311	0.162	0%	0%
M4 Filter + rating	**0.941±0.020**	**0.821±0.019**	**0.526**	**0.249**	0%	0%
M5 Filter + v1	0.906±0.013	0.809±0.013	0.384	0.239	0%	0%
M6 Filter + v2	0.900±0.013	0.784±0.022	0.362	0.206	0%	0%
Violation floor					28.7%	9.5%

M6 is the production method. Violation floor: violation rate of the oracle when it must always return five trainers.

Validity. Hard filtering removes infeasible trainers (Table III): the violation rate among the top five falls from about 95% for random and rating-only ordering to 0% for every filtered method. Scoring without a filter (M2) still yields 49.6% violations in S1 and 37.8% in S3, clearly above the floors of 28.7% and 9.5%. The effect of filtering on ranking, M6 − M2, ranges from +0.25 to +0.30 NDCG@5, with every interval above zero.

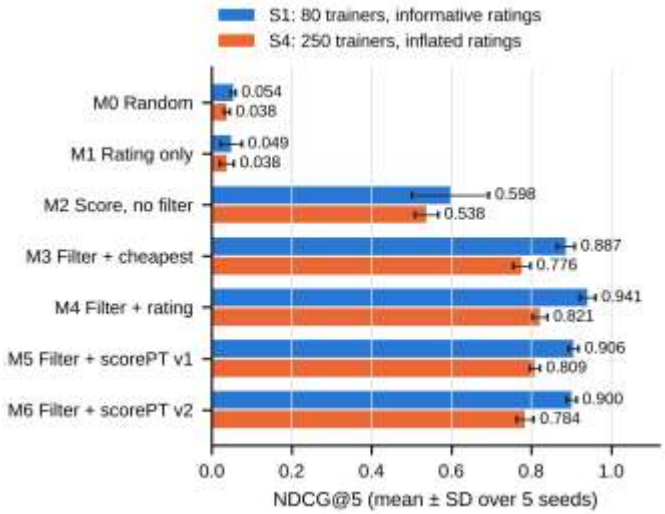


Fig. 2. NDCG@5 of the main methods in scenarios S1 and S4 (five seeds × 300 clients; bars show ±1 SD).

Ranking quality. After filtering, the production score M6 ranks below the filter-then-rating baseline M4 in all four scenarios (Fig. 2). The paired differences M4 − M6 are +0.041 [0.029, 0.052] in S1, +0.022 [0.002, 0.041] in S2, +0.075 [0.063, 0.087] in S3 and +0.037 [0.018, 0.056] in S4, the S2 interval lying close to zero. The ablation in Table IV points to three measurable causes. First, criteria already enforced by the filter are scored again and become constant, so removing goal, schedule and budget (A1) leaves NDCG@5 essentially unchanged. Second, integer rounding produces ties in 69% to 89% of top-five lists. Third, body-metric evidence only counts similar past clients and applies to 19% to 28% of candidates; removing it (A3) increases NDCG@5. Reputation, in contrast, contributes: removing it (A2) lowers both NDCG@5 and HitRate@1. Within this simulation, and under its assumptions about latent quality, scorePT v2 therefore does not outperform filtering followed by sorting by rating, and the body-metric component, in its current form, does not yet contribute to ranking.

TABLE IV
ABLATION OF SCOREPT V2 IN SCENARIO S3

Variant	NDCG@5	Hit@1	Ties@5
M6 v2, full	0.810±0.016	0.233	85.7%
A1 without goal, schedule, budget	0.812±0.013	0.246	59.0%
A2 without reputation	0.784±0.024	0.178	96.8%
A3 without body-metric evidence	**0.850±0.013**	**0.295**	93.1%
X2 unrounded, rating tie-break	0.815±0.012	0.240	59.5%

Ties@5: share of top-five lists with at least two trainers that contain two or more trainers with equal score.

B. Experiment C: The Gymini Pipeline on the Benchmark

Table V reports the deterministic pipelines together with the LLM configurations. The current pipeline G0 fills 57.8% (52/90) of the top-three positions, but only 26.9% of the returned positions are valid; its Top-1 validity is 33.3% and its case success 36.7%, and 4 clients receive an empty list although a valid trainer exists. Random selection R reaches a Valid@3 of 8.8% and a case success of 24.7%. Of the 38 invalid positions of G0, 25 violate the district rule (H5), 16 have no time overlap of at least 60 min (H2) and 7 lack a required specialty (H1),

while none violates the budget or mode rules (one position can violate several rules). The cause lies in the data model: Gymini stores availability only by day and location only by province, and supports only four goals. The four false empty lists (C02, C14, C17, C29) occur because G0 requires availability on every selected day or because the goal cannot be mapped. Adding hour, district and per-session price constraints (G1) reaches 100% on every validity metric with the maximum coverage of 48/90; this holds by construction, since the filter is built from H1–H5, and shows which constraints G0 lacks rather than ranking accuracy. These figures use the behavior in which a budget is not required; on the measured commit, the three clients without a budget limit fail with an HTTP 422 error, which lowers the case success of G0 to 33.3%.

C. Experiment B: Real LLM in Three Configurations

TABLE V
VALIDITY ON THE BENCHMARK, 30 CLIENTS (%, 95% WILSON CI)

	Pos.	Valid@3	Top-1	Case succ.	Empty†	Recall
R random 3/20	—	8.8	8.8	24.7	0	0.161
G0 (no LLM)	52	26.9 [16.8–40.3]	33.3 [19.2–51.2]	36.7 [21.9–54.5]	4	0.272
L1	17	100 [81.6–100]	46.7 [30.2–63.9]	46.7 [30.2–63.9]	16	0.317
L0	48	100 [92.6–100]	100 [88.6–100]	100 [88.6–100]	0	1.000
G1 (no LLM)	48	100*	100*	100*	0	1.000
L2	48	100** [92.6–100]	100** [88.6–100]	100** [88.6–100]	0	1.000

Pos.: positions returned (maximum possible valid positions 48). †Empty answer although a valid trainer exists. *By definition: the G1 filter is built from H1–H5. Largely by design: L2 receives exactly the valid set. Fabricated, out-of-list or duplicate trainer codes: 0/0/0 for L0, L1 and L2*

Validity. L0, which inspects all 20 trainers, returned 48 positions; no recommendation that violates a constraint was observed in the 30 test cases (95% CI of case success 88.6% to 100%), and no fabricated, out-of-list or duplicate trainer code occurred (95% CI of the fabrication rate 0% to 7.4%). This is a result within the scope of the benchmark, not an accuracy guarantee in deployment. L2 also reached 100%, but largely by design: its candidate list equals the valid set and 27 of 30 clients have at most three valid trainers, so returning the whole list gives a recall of 1. Placing the LLM after the weak current filter (L1) removed invalid recommendations, raising Valid@3 from 26.9% to 100% [81.6–100], but the model could not recover trainers already dropped by the filter: 16 of 30 clients received an empty answer although a valid trainer existed, and case success rose from 36.7% to 46.7% with overlapping intervals.

TABLE VI
NDCG@3 ON THE 12 CLIENTS WITH AT LEAST TWO VALID TRAINERS

	NDCG@3 [95% CI]	Diff. vs. random valid [95% CI]	Best at rank 1
Worst order of valid trainers	0.870	—	—
Random valid ranking	0.935 (SD 0.019)	—	—
G1 scorePT	0.950 [0.899, 0.989]	+0.014 [−0.030, +0.060]	8/12
L2 full + LLM	0.965 [0.939, 0.989]	+0.029 [−0.0001, +0.062]	8/12
L0 LLM only	0.969 [0.943, 0.991]	+0.034 [−0.005, +0.076]	9/12
L1 Gymini + LLM	0.460 [0.280, 0.632]	−0.475 [−0.664, −0.288]	6/12
G0 current Gymini	0.286 [0.103, 0.486]	−0.649 [−0.848, −0.426]	3/12

Random valid ranking: 1000 seeds; 95% of seeds lie in 0.897–0.970.

Ranking quality. On the 12 clients with at least two valid trainers (Table VI), G1, L2 and L0 reach NDCG@3 of 0.950, 0.965 and 0.969, against 0.286 for G0 and 0.460 for L1. Randomly ordering only the valid trainers, however, already gives 0.935, and even the worst possible order of valid trainers gives 0.870. None of G1, L2 and L0 differs significantly from the random valid ranking, and the paired differences L0 − L2 = +0.004 [−0.023, 0.033] and L2 − G1 = +0.015 [−0.004, 0.047] are not significant either. The soft labels are coarse: 8 of the 12 clients have tied labels, 3 (C03, C06, C21) have all valid trainers tied, so that every order yields NDCG = 1, and 9 have at most three valid trainers. This is a ceiling effect: on this benchmark NDCG@3 mainly measures whether only valid trainers are selected, and the data do not allow us to conclude that any of G1, L2 or L0 orders valid trainers better than another or better than random.

TABLE VII
ACCURACY OF EXPLANATIONS

	L0 free	L1 free	L2 free	Claim catalog
Explanations	48	17	48	27 (168 sent.)
Numeric facts correct	48/48	17/17	48/48	124/156 sent. fully correct; 0 wrong numbers
(a) Overclaim (written by LLM)	1/48, 2.1% [0.4–10.9]	6/17, 35.3% [17.3–58.7]	20/48, 41.7% [28.8–55.7]	**0/27** [0–12.5]
(b) Unsupported "verified" (from template)	0/48	0/17	0/48	**23/27, 85.2%** [67.5–94.1]
Schedule sentence above hour overlap	—	—	—	7/27, 25.9% [13.2–44.7]
Invalid or foreign claim IDs	—	—	—	0/130 [0–2.9]

Explanation accuracy. Free-text explanations reproduced every numeric fact correctly (Table VII). Their errors are of another kind: 27 of 113 explanations (23.9% [17.0–32.5]) turn the simulated coaching traits of a trainer into statements that the trainer is rated or reviewed as having a quality, i.e., type (a) overclaims, all confirmed by manual reading. The field name coaching_feedback_traits, which contains the word feedback, may contribute to this wording. The claim catalog eliminates type (a): the LLM writes no sentences, none of its 130 selected claim identifiers was invalid or belonged to another trainer, and the fallback was never used. Its fixed experience template, however, always states that the years of experience come from a verified profile, although these values are simulated and unverified, so 23 of 27 explanations (85.2% [67.5–94.1]) contain a type (b) unsupported verification claim (Fig. 3). In addition, 7 of 27 schedule sentences are correct by day but higher than the actual hour overlap. Overall, 124 of 156 factual sentences are fully correct and none contains a wrong number. The claim catalog thus controls content added by the LLM but does not by itself ensure that the wording matches the verification level of the source; its error is systematic and deterministic and can be removed by changing the template without re-running the LLM, whereas the errors of free text occur at random.

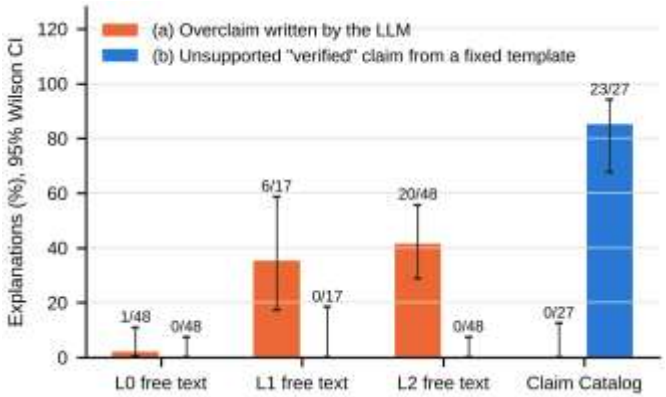


Fig. 3. Unsupported statements in explanations: overclaims written by the LLM (a) and unsupported "verified" claims produced by a fixed template (b). Bars show 95% Wilson intervals.

Stability and cost. For the 10 repeated clients, the top three was identical including its order in 9 of 10 runs for L0 [59.6–98.2%] and in 10 of 10 for L1 and L2 [72.2–100%], with a mean Jaccard index of 1.00 in all cases; the intervals overlap, so stability cannot be distinguished. Per call, L0 used 11,471 input and 780 output tokens (344 of them reasoning), L1 2,991 and 481 (172), and L2 2,841 and 725 (287), with median latencies of 5.7 s, 4.5 s and 5.3 s. The cost per 1000 calls is about 61 USD for L0, 22 USD for L1, 26 USD for L2 and 11 USD for the claim catalog. Under these test conditions, L2 matches the validity of L0 at about 57% lower cost and with about one quarter of its input tokens, bearing in mind that the validity of L2 is largely a consequence of its design.

V. Discussion

What the data support. Hard filtering prevents recommendations that violate mandatory requirements, and the granularity of the constraints matters as much as their presence: day-level availability and province-level location made about 73% of the positions returned by the current pipeline invalid. An LLM placed after a weak filter removes invalid recommendations but cannot restore trainers that the filter dropped. Free-text explanations overstate simulated traits in about one explanation in four, while the claim catalog removes content added by the LLM.

What holds only under conditions. That scorePT v2 does not outperform filtering followed by rating, and that the count-based body-metric evidence does not contribute, holds within the simulation and depends on its assumptions about latent quality. That L0 produced no violating recommendation, and that L2 reached the same validity at about 57% lower cost with about one quarter of the input tokens, holds on a small benchmark of structured data; the validity of L2 is largely a consequence of its design.

What the data do not show. The benchmark cannot tell whether L2, L0 or G1 orders valid trainers better than one another or better than random, whether L2 is more stable than L0, whether filtering before the LLM is necessary when hundreds of trainers are available, or whether the model correctly answers that no valid trainer exists. These remain open questions.

Implications for the system. Three changes follow directly from the results. Hard constraints should be checked deterministically at the level of hours and districts before the LLM. The score should not re-score criteria that the filter already enforces and should avoid integer rounding, which produces ties; whether such changes improve ranking has to be tested on a benchmark that can measure ranking. Explanation templates should state the provenance of each attribute, for example "according to profile data" for an unverified value, and data fields should not be named in a way that suggests client feedback when they are simulated.

Limitations. The operational attributes of the trainers, all clients and the whole of Experiment A are synthetic, and the specialty labels match public statements exactly for only 11 of 20 trainers. The synthetic benchmark data were generated with Claude, and the evaluated model is also a Claude model; this shared origin may affect the results in a direction we cannot estimate, which is a further reason to repeat the study with independently produced data. The benchmark is small and structured: 18 of 30 clients have a single valid trainer, no client has zero, the prompt states the constraints explicitly, and near-threshold schedules or prices, missing or contradictory data, vague natural-language requests and prompts with hundreds of trainers were not tested. Only one LLM was used, each configuration was run once (twice for 10 clients), the soft labels are rule-based with three levels and many ties, the conversational tier T1 was not evaluated, and there is no user study. The 100% rates therefore reflect both the methods and the ease of the test data.

VI. Conclusion

This paper evaluated LLM-driven personal trainer matching in Gymini along three separate aspects, using real public trainer profiles, AI-generated synthetic attributes and clients declared field by field, and a code-generated simulation. Deterministic hard filtering was the component that most clearly improved validity, and the current Gymini data model, with day-level availability and province-level location, was the main source of invalid recommendations. On the benchmark, the LLM alone made no constraint-violating recommendation, and combining it with full filtering and scoring reached the same validity at lower cost, but ranking quality could not be distinguished from random ordering of valid trainers, and neither free-text explanations nor the claim catalog fully respected the verification level of the data. Future work will build a harder benchmark with cases that have no valid trainer, near-threshold constraints and natural-language requests, obtain ranking labels from expert trainers, repeat the runs with several models, correct the claim templates and move the data model to hourly availability and district-level location.

Acknowledgment

Claude Opus 5.5 (Anthropic) [10] was the model evaluated in Experiment B. Claude was also used to generate the synthetic trainer attributes and the 30 client scenarios, which the authors reviewed and locked before the experiments, and to assist in

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